

AR Wardrobe Trial

A 2D Camera-Based Virtual Dressing Room Using Pose Estimation and Person Segmentation

1. Introduction

The AR Wardrobe Trial is a 2D camera-based virtual dressing room that allows users to try different outfits virtually without physically wearing them. The system uses a webcam or mobile camera to capture the user and computer-vision techniques to detect their body position and identify the person within the camera frame.

The proposed system combines MediaPipe Pose for detecting body landmarks such as shoulders, elbows and hips with person segmentation for identifying the user's body region and improving the visual layering of virtual clothing.

The selected clothing items are represented as transparent 2D images and are dynamically resized, positioned and overlaid according to the user's detected body measurements and movements.

2. Problem Statement

Traditional online clothing shopping does not allow users to physically see how an outfit may look on their body before purchasing it. This can lead to uncertainty regarding appearance, size and fit, as well as increased product returns.

Physical trial rooms, on the other hand, require users to visit a store and manually change outfits.

Therefore, there is a need for an accessible virtual dressing system that allows users to:

- Try different outfits without physically changing clothes.
 - View clothing in real time through a camera.
 - Adjust virtual clothing according to body position.
 - Receive an estimated size recommendation.
 - Evaluate whether an outfit appears appropriately fitted.
 - Interact with the system using simple gestures.
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3. Proposed Solution

The proposed system will use a webcam to capture the user's live video. The system will process each frame using two major computer-vision components:

MediaPipe Pose

Detects important body landmarks such as:

- Left and right shoulders
- Elbows
- Wrists
- Left and right hips
- Knees
- Ankles

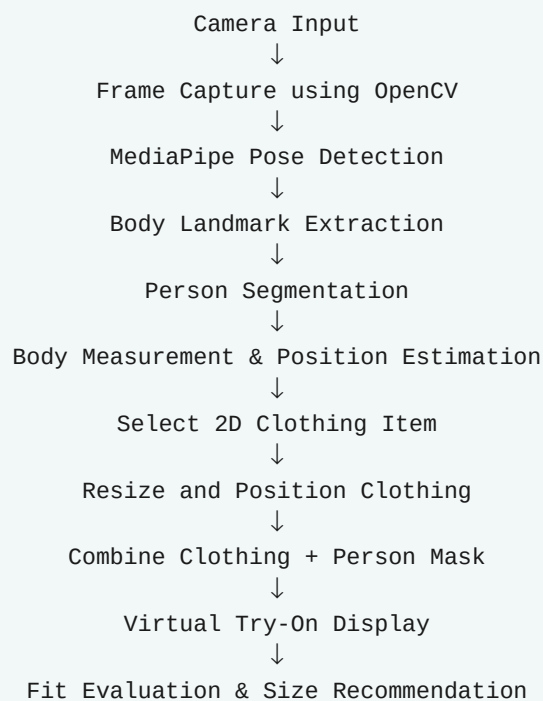
These landmarks will be used to estimate the position, orientation and approximate dimensions of the user's body.

Person Segmentation

Creates a mask separating the user from the background. This information will be used to improve the placement and layering of virtual clothes so that the outfit appears more naturally integrated with the user's body.

The detected information will then be combined with transparent 2D clothing images to create the virtual try-on experience.

4. System Workflow



5. Main Features

5.1 Real-Time Virtual Outfit Trial

Users can select clothing items such as:

- T-shirts
- Shirts
- Tops
- Jackets
- Dresses

The selected clothing will be displayed over the user's body in real time.

5.2 Pose-Based Clothing Placement

The system will use shoulder and hip landmarks to determine where the clothing should be positioned. For example: shoulder distance determines clothing width, while shoulder/hip position determines clothing placement. As the user moves closer, farther away or changes position, the virtual clothing will adjust accordingly.

5.3 Person Segmentation

Segmentation will identify the user's body region separately from the background. This can improve the visual realism of the virtual outfit by allowing appropriate foreground/background layering.

5.4 Size Recommendation

The system can estimate body proportions using detected landmarks and compare them with predefined clothing-size ranges (estimated measurements → size rules → recommended size). The recommendation will be presented as an approximate suggestion rather than an exact physical measurement.

5.5 Virtual Fit Evaluation

The system will compare the estimated body dimensions with the dimensions associated with the selected clothing, and can provide a simple result such as:

- Good Fit
- Slightly Loose
- Slightly Tight

This provides additional functionality beyond simply displaying a clothing image.

5.6 Gesture-Based Interaction

Hand/pose landmarks can be used to implement simple touchless controls, for example:

- Hand movement → browse outfits
- Hand gesture → select outfit
- Another gesture → change clothing
- Exit gesture → close trial mode

This allows the system to be operated without touching the screen.

6. Technologies Required

Software

- Python
- OpenCV — camera input and image processing
- MediaPipe — pose/landmark detection and segmentation
- NumPy — numerical and image-processing operations
- Tkinter / Flask / Streamlit — optional user interface
- Pillow (PIL) — handling and manipulating clothing images

Hardware

- Laptop/PC
 - Webcam or built-in camera
 - Optional mobile camera for future deployment
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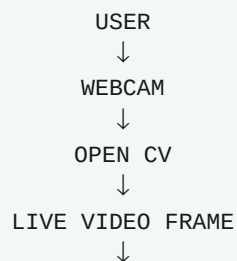
7. Clothing Dataset

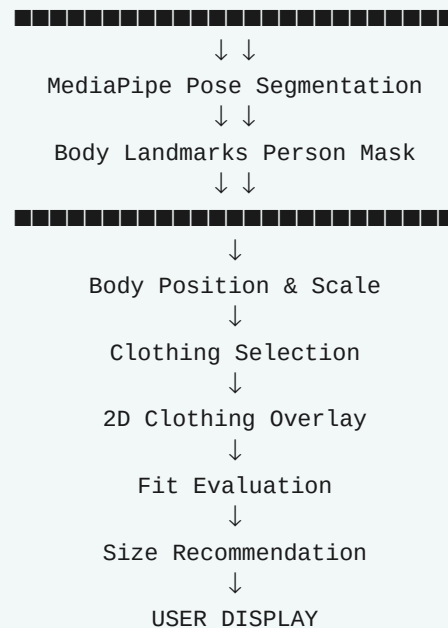
The system will use a collection of transparent-background 2D clothing images. Each clothing item can contain metadata such as:

Clothing	Type	Approx. Size Range	Category
Shirt 1	Shirt	S–M	Casual
Shirt 2	Shirt	M–L	Formal
Top 1	Top	S–M	Casual
Dress 1	Dress	M–L	Party

The clothing images will preferably be PNG files with transparent backgrounds so that they can be overlaid on the camera frame.

8. System Architecture





9. Expected Outcome

The final system will provide a working 2D virtual dressing room where a user can stand in front of a camera and virtually try different clothing items. The system is expected to:

- Detect the user's body in real time.
- Track body movement using pose landmarks.
- Identify the person using segmentation.
- Position and scale virtual clothes according to the user's body.
- Allow users to switch between outfits.
- Provide an approximate size recommendation.
- Provide a basic virtual fit evaluation.
- Support touchless interaction through gestures.

10. Unique Selling Point

The major USP of the proposed project is that it is not limited to simply placing a clothing image over a person. The system combines real-time pose tracking, person segmentation, virtual outfit overlay, size recommendation, fit evaluation and gesture-based interaction in a single lightweight 2D camera-based system.

This makes the project practical for online shopping applications while avoiding the computational complexity of a full 3D virtual try-on system.

11. Scope

Current Scope

The initial system will focus on:

- Upper-body clothing.
- Front-facing camera input.
- 2D transparent clothing assets.
- Real-time pose tracking.
- Person segmentation.
- Basic size recommendation.
- Basic fit evaluation.
- Gesture-based outfit selection.

Future Scope

The project can later be extended to:

- Full-body clothing.
 - Pants and skirts.
 - More accurate body measurements.
 - Improved cloth deformation.
 - AI-based outfit recommendations.
 - Multiple body poses.
 - Mobile application deployment.
 - Personalized fashion recommendations.
 - Integration with e-commerce platforms.
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12. Limitations

Since this is a 2D virtual try-on system, it will not reproduce the complete physical behavior of real fabric.

The accuracy can also depend on:

- Camera quality.
- Lighting conditions.
- User's distance from the camera.
- Body orientation.
- Quality of clothing images.
- Accuracy of the estimated measurements.

Therefore, the size and fit results should be treated as recommendations/estimates rather than guaranteed physical fit.

13. Conclusion

The AR Wardrobe Trial proposes a practical 2D virtual dressing-room solution using computer vision. MediaPipe Pose will provide body landmarks for determining the position and scale of virtual clothing, while person segmentation will provide information about the user's body region for improved visual compositing.

By combining virtual outfit trial, approximate size recommendation, fit evaluation and gesture-based interaction, the project aims to provide a more interactive and useful shopping experience than a basic clothing-overlay application.

The project demonstrates the practical application of computer vision, image processing, augmented reality concepts and human-computer interaction in the fashion and e-commerce domain.